LI-CHUN LU

Li-Chun Lu | lichunlu@media.mit.edu | Google Scholar

EDUCATION

National Taiwan University (NTU) **Taipei, Taiwan**

Doctoral Student in Graduate Institute of Communication Engineering, Data Science & Smart Networking Group *Sep. 2025 - Present*

- Research areas: Large language model (LLM); human–AI interaction & augmentation of human agency, cognition, and creativity.

B.S. in Electrical Engineering *Sep. 2020 - Jun. 2025*

- Selected Courses: Algorithms, Computer Architecture, Computer Programming, Intro. to Generative AI, VR Game Programming.

National Cheng Kung University (NCKU) **Tainan, Taiwan**

Bachelor of Architecture *Sep. 2019 - Aug. 2020 (Transferred)*

RESEARCH EXPERIENCES

Massachusetts Institute of Technology (MIT) Media Lab, Fluid Interfaces Group **Massachusetts, United States**

Visiting Student Researcher | Advisor: Prof. Pattie Maes *May 2026 – Present*

- Lead technical and methodological development for a project in cognitive offloading and automation, including constructing taxonomies, analyzing real-world chatbot interactions, and designing human studies and automated-classification pipelines.

NTU Robot Learning Lab **Taipei, Taiwan**

Research Assistant, Student Researcher | Advisor: Prof. Shao-Hua Sun *Oct. 2023 – Present*

- Conducted research on creativity in LLM agents, including designing a role-playing discussion framework for multi-agent systems that improved creative performance by 20%, and collaborating on a persona-blending framework that further reduced costs.
- Devised literature-based evaluations for creativity and verified through 1,400+ human annotations across four benchmarks.

University of California, Los Angeles (UCLA) Peng's Language Understanding & Synthesis Lab **California, United States**

Visiting Student Researcher | Advisor: Prof. Nanyun (Violet) Peng *Nov. 2024 – Apr. 2025*

- Led a project analyzing four creativity evaluation metrics across three domains: creative writing, problem-solving, and research ideation, curated or verified datasets for each, and identified cross-metric inconsistencies.
- Diagnosed key limitations of each metric, including lexical bias in the creativity index, sensitivity to overconfidence in perplexity, inability to capture conceptual-level creativity in syntactic templates, and instability and bias in LLM-as-a-Judge.

NTU Deep Learning & Human Language Processing Lab **Taipei, Taiwan**

Student Researcher | Advisor: Prof. Hung-yi Lee *Jul. 2023 – Oct. 2024*

- Proposed and processed a dataset, "Third Tone Sandhi Recognition in Mandarin," for Dynamic-SUPERB to evaluate universal speech models, based on the NCCU Corpus of Spoken Taiwan Mandarin dataset.
- Applied the Direct Preference Optimization (DPO) framework in reinforcement learning to train a text-instruction-guided voice conversion model, collaborating with two teammates to enhance its performance.

NTU Human Computer Interaction Lab **Taipei, Taiwan**

Student Researcher | Advisor: Prof. Mike Y. Chen *Aug. 2023 – May. 2024*

- Demonstrated and presented our game at the Student Game Competition (SGC). Implemented three levels of sensation substitution and remapping in a virtual reality game using Unity to enhance human comprehension of animal senses.
- Surveyed prior works, collected, and analyzed Mass Rapid Transit speed data to justify the validation of paper hypotheses and questionnaire design; designed figures and edited a demo video to visualize paper information.

PUBLICATIONS

- **Li-Chun Lu**, Miri Liu, Pin-Chun Lu, Yufei Tian, Shao-Hua Sun, Nanyun Peng, "Rethinking Creativity Evaluation: A Critical Analysis of Existing Creativity Evaluations," *Proceedings of the European Chapter of the Association for Computational Linguistics*, 2026.
- Tsung-Min Pai, Jui-I Wang, **Li-Chun Lu**, Shao-Hua Sun, Hung-Yi Lee, Kai-Wei Chang, "BILLY: Steering Large Language Models via Merging Persona Vectors for Creative Generation," *Proceedings of the European Chapter of the Association for Computational*

Linguistics, 2026. **Oral.**

- Chien-yu Huang et al. (including Li-Chun Lu), "Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Measuring the Capabilities of Spoken Language Models with 180 Tasks," *Proceedings of the Thirteenth International Conference on Learning Representations*, 2025.
- **Li-Chun Lu***, Shou-Jen Chen*, Tsung-Min Pai, Chan-Hung Yu, Hung-yi Lee, Shao-Hua Sun, "LLM Discussion: Enhancing the Creativity of Large Language Models via Discussion Framework and Role-Play," *Proceedings of the Conference On Language Modeling*, 2024.
- Yu Lun Hsu, Chien-Ting Lu, **Li-Chun Lu**, Chih-Heng Tam, Yu-Chieh Sun, Ting-Kang Wang, "AnimalSense: Understanding Beyond-human Sensory Capabilities of Animals via VR Games," *Extended Abstracts of the Conference on Human Factors in Computing Systems*, 2024. **Runner-Up Award.**
- Pin-Chun Lu et al. (including Li-Chun Lu), "VeeR: Exploring the Feasibility of Deliberately Designing VR Motion that Diverges from Mundane, Everyday Physical Motion to Create More Entertaining VR Experiences," *Proceedings of the Conference on Human Factors in Computing Systems*, 2024.

TEACHING EXPERIENCES

Seminar, NTU — **Taipei, Taiwan**

Teaching Assistant (TA) — *Sep. 2025 – Jun. 2026*

- Coordinate with speakers, assist with hosting talks and Q&A sessions, and manage event logistics and administrative procedures.

Intro. to Generative Artificial Intelligence Course, NTU — **Taipei, Taiwan**

Teaching Assistant — *Feb. 2024 – Jun. 2024*

- Handled questions during TA hours and via email from 1,000 students and 1,000 auditors, in collaboration with 37 other TAs.
- Designed slides as teaching materials for an assignment, along with an LLM essay evaluation system, helping students practice and understand the concept of prompting, in collaboration with two other TAs.

GRANTS & AWARDS

Scholarship, *Y.L. Lin Program, NTU* — *Jun. 2024*

Runner-Up Award, *Student Games Competition, CHI 2024* — *May 2024*

Excellence Award, *Architecture Video Storytelling Project, Dept. of Architecture, NCKU* — *Feb. 2020*

INTERNSHIPS

NTU Y.L. Lin Program, Taiwan FactCheck Center — **Taipei, Taiwan**

Collaboration Intern — *Dec. 2023 – May 2024*

- Designed an AI Media Literacy Web Game to address misinformation and enhance public understanding of AI.
- Conducted user research via individual interviews with 15 people to improve the interface design, gameflow, and effectiveness.

Solution Engineer Team, Synopsys Taiwan Co. — **Hsinchu, Taiwan**

Technical-Engineering Intern — *Jul. 2023 – Aug. 2023*

- Debugged four sets of Process Design Kits (PDKs) with a teammate, each encompassing both Schematic-versus-Schematic (SvS) and Layout-versus-Layout (LvL) analyses; utilized Python, TCL, and Custom Compiler to inspect the callback functions of PDKs.

ACADEMIC SERVICES

Reviewer, *ICLR 2026; CHI 2026 (Poster); IEEE Spoken Language Technology Workshop 2024*

Volunteer, *COLM DEI Scholarship Program 2024; Conference on Computational Linguistics and Speech Processing (ROCLING) 2025*

SIDE PROJECTS

Smart Driving Assistance, *Final Project, Embedded Systems Course, NTU* — *Dec. 2023*

Emotional Ocean: an Interactive Device to Raise Awareness of Emotional Issues, *TAICHI Demos Session, CSCW* — *Nov. 2022*